- 1117838570 2005.06.03 R02-M1-N0-C:J12-U11 RAS KERNEL INFO instruction cache parity error corrected

Log Intelligence Agent

I wanted the same logs to give an engineer, a team lead and a manager the answer each of them actually needs.

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Team ultraviolet-owls · SereneDB challenge · National AI Hackathon 2026

github.com/Jay-0903/log-intelligence-agent

4,747,963 BGL lines

207,820 OpenStack lines

16 tools

Nobody reads five million log lines.

The answers are in there. But an AI can’t fit them in a prompt, and three different people are asking three different things:

Engineer, on call	“What broke, and what happened right before?”
Team lead	“Is this a pattern or one bad day? And whose is it?”
Manager	“What do we fix first?”

A shorter answer isn’t the manager’s answer.

So I let the database do the reading.

question → Claude picks a tool → SereneDB searches or counts → answer + the SQL it ran

Four rules I set myself, each because of something that could go wrong:

1 **The database counts; the model explains.**

An AI can't read 4.7M lines, and counts have to be exact, not estimated.

2 **No free-form SQL.**

A made-up query can be quietly wrong. Tools are tested and bounded.

3 **Every answer shows its SQL.**

Someone is going to act on these numbers.

4 **Facts and guesses are labelled.**

Logs show what happened, not why.

Each person got their own tools.

Who	Tools	Why they need them
Engineer	search_logs_fulltext · get_nearby_events · request_trace	the exact lines and the minutes around them
Team lead	alert_profile · alert_categories · count_alerts_by_team · trend_by_day · trend_by_week · top_nodes_for_alert	spread over days separates bursts from recurring problems
Manager	openstack_by_component · latency_stats · alert_profile	rates and owners, to rank what matters
Everyone	corpus_scope · team_ownership	what data an answer covers, and who acts on it
Bonus	otel_setup · otel_aggregate · otel_search	checking it scales to a billion rows

Claude picks these on its own. I never tell it which one.

Engineer · live in Claude

One bad chip took down a whole job.

What I asked

“Investigate KERNRTSP alerts: which nodes and days, example lines, and what happened before and after the most recent one?”

3,983 lines on 3,394 nodes over 126 days. Half of them from one day in June 2005, then a steady trickle.

```
3 Jan 2006 · node R27-M0-NA-C:J11-U11
02:24–03:24 nothing. an hour of quiet.
03:24:46 L2 dcache parity machine check
03:25:05 rts panic! - stopping execution
03:25:06 link severed: I/O nodes in six racks
```

Likely cause, still to be confirmed: a hardware fault in that chip’s cache ended one large job.

Next: diagnose R27-M0-NA-C:J11-U11. R63-M0-NE-C:J12-U01 has the most panics of any node (22), so it’s the first to replace.

Full BGL data, 4,747,963 lines · BM25 search with a time filter, then events around that moment on the same node

The loudest alert wasn't the one that kept coming back.

Alert	Lines	Nodes	Days	Peak day	Owner
KERNDTLB	152,734	541	10	99.6%	Kernel Platform
KERNSTOR	63,491	22,529	4	100%	Kernel Platform
APPSEV	49,651	1,024	70	6.2%	App Runtime
KERNMNTF	31,531	1,024	26	13.8%	Kernel Platform
KERNRTSP	3,983	3,394	126	51.4%	Kernel Platform
KERNMC	342	265	144	2.9%	Kernel Platform

6 of 41 alert types. **Red**: a burst (90%+ on one day). **Blue**: still showing up at the end of the data.
Whole BGL corpus, Jun 2005 – Jan 2006 · alert_profile, count_alerts_by_team, trend_by_day

99.6%

of KERNDTLB's 152,734 lines happened on one day, 12 June 2005. One bad day, not a recurring defect.

KERNMC: only 342 lines, but on 144 days. Easy to miss.

What I’d fix first.

1	**Application Runtime**	APPSEV: 70 days, all 1,024 nodes, no day above 6%. Something systemic, not one bad machine. Start with what changed in mid-November.
2	**API Gateway**	nova-api: a steady 2.01% of lines are errors, every day (2,068 of 102,868). Find out what that 2% is, then look at the slow tail (p99 514 ms).
3	**Kernel Platform**	The quiet, long-running ones: KERNMC (144 days) and KERNRTSP (126 days). Check the nodes that show up in both.
4	**Compute**	nova-compute: 204 error lines (0.20%), one repeating image-cache job. Low priority.
↺	**Look back, don’t chase**	One-day bursts like KERNDTLB and KERNSTOR become post-incident reviews, not ongoing work.

Ranked by how long a problem lasts, not how loud it is · owners from my ownership table · no made-up costs or downtime

Every number comes with the query behind it.

```
[scope: BGL, 4,747,963 lines | 2006-01-03 to 2006-01-04 | BM25
full-text]

SELECT …, BM25(bgl_idx.tableoid) AS score
FROM bgl_idx
WHERE (message @@ ts_phrase('rts')
   OR message @@ ts_phrase('panic'))
  AND ts BETWEEN … ORDER BY score DESC

→ R27-M0-NA-C:J11-U11 · rts panic! · score 8.382
```

The bug that taught me why this matters

Halfway through, search on the full data quietly fell back to plain text matching.

Because every tool says *why* it falls back, I could rule out the data and the index.

It was how the database driver sent parameters. One line fixed it.

Without that, I'd have shipped wrong answers and never known.

What this can and can’t tell you.

It can tell you	It can’t tell you
exact line counts over all the data	the confirmed root cause
how problems spread over nodes and days	how many incidents (it counts lines)
which team owns each component	downtime, cost or customer impact
what happened around an event	labelled anomalies for OpenStack

Question → evidence → recommendation, for each person. Every number checkable.

Bonus: 1 billion OpenTelemetry rows, queried straight from S3 with read_parquet. 12–16 s per aggregation, nothing downloaded.

Thanks for watching! – Jay